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Studierichting: Informatica
Oefeningenreeks 3E nr. 11

Opgave: $l(x) = \dfrac{k\cos x}{d^2}$

$\overset{SOS}{\Rightarrow} d = \dfrac{a}{\sin x}$

$$l(x) = \frac{k\cos x}{\left(\dfrac{a}{\sin x}\right)^2} = \frac{k\sin^2 x\cos x}{a^2}$$

$$l'(x) = \frac{k}{a^2}\left(\sin 2x\cos x - \sin^3 x\right) = \frac{k}{a^2}\left(2\sin x\cos^2 x - \sin^3 x\right) = \frac{k}{a^2}\sin x\left(2\cos^2 x - \sin^2 x\right)$$

$$l'(x) = 0 \Rightarrow \frac{k}{a^2}\sin x\left(2\cos^2 x - \sin^2 x\right) = 0$$

$\Rightarrow k = 0$ of $\sin x = 0$ of $2\cos^2 x - \sin^2 x = 0$

$\Rightarrow k$ gegeven

$\Rightarrow \sin x = 0 \Rightarrow x = 2m\pi$ of $x = \pi + 2m\pi$ met $\pi \in \mathbb{Z} \Rightarrow L$ naar boven of onder

$\Rightarrow 2\cos^2 x - \sin^2 x = 0 \Rightarrow 2\cos^2 x - (1 - \cos^2 x) = 0 \Rightarrow 2\cos^2 x - 1 + \cos^2 x = 0$

$\Rightarrow 3\cos^2 x = 1 \Rightarrow \cos^2 x = \dfrac{1}{3} \Rightarrow \cos x = \sqrt{\dfrac{1}{3}} \Rightarrow \cos x = \pm\dfrac{\sqrt{3}}{3}$

$\Rightarrow \cos x = -\dfrac{\sqrt{3}}{3} \Rightarrow$ niet mogelijk

$\Rightarrow \cos x = \dfrac{\sqrt{3}}{3} \Rightarrow x = \mathrm{Bgcos}\,\dfrac{\sqrt{3}}{3} + 2m\pi$ met $m \in \mathbb{Z}$

mogelijke x-waarden: $x = \mathrm{Bgcos}\,\dfrac{\sqrt{3}}{3} + 2m\pi$ met $m \in \mathbb{Z}$

$$l''(x) = \frac{k}{a^2}\left((2\cos x\cos 2x - \sin x\sin 2x) - \left(3\sin^2 x\cos x\right)\right) = \frac{k}{a^2}\left(\cos x\left(2\cos 2x - 3\sin^2 x\right) - \sin x\sin 2x\right)$$

$$l''\left(\mathrm{Bgcos}\,\frac{\sqrt{3}}{3}\right) \approx -0,57073\frac{k}{a^2} < 0 \Rightarrow \text{maximum}$$

$$d^2 = a^2 + h^2 \Rightarrow h^2 = d^2 - a^2 \Rightarrow h = \sqrt{d^2 - a^2} = \sqrt{\left(\frac{a}{\sin x}\right)^2 - a^2}$$

$$h = \sqrt{\frac{a^2}{\sin^2 x} - a^2} = \sqrt{\frac{a^2 - a^2\sin^2 x}{\sin^2 x}} = \sqrt{\frac{a^2(1 - \sin^2 x)}{\sin^2 x}} = \sqrt{\frac{a^2\cos^2 x}{\sin^2 x}}$$

$$h = \sqrt{a^2\cot^2 x} = a\cot x = a\cot\left(\mathrm{Bgcos}\,\frac{\sqrt{3}}{3}\right) = a\frac{\cos\left(\mathrm{Bgcos}\,\frac{\sqrt{3}}{3}\right)}{\sin\left(\mathrm{Bgcos}\,\frac{\sqrt{3}}{3}\right)}$$

$$h = a\frac{\frac{\sqrt{3}}{3}}{\sqrt{1-\frac{\sqrt{3}^2}{3^2}}} = a\frac{\frac{\sqrt{3}}{3}}{\sqrt{1-\frac{1}{3}}} = a\frac{\frac{\sqrt{3}}{3}}{\frac{\sqrt{2}}{\sqrt{3}}} = a\frac{\sqrt{3}}{3}\frac{\sqrt{3}}{\sqrt{2}} = a\frac{3}{3\sqrt{2}} = \frac{a}{\sqrt{2}}$$

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References